

William Abbruzzese

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Education

Colorado College, Bachelor's in Physics

May 2028

GPA: 3.97/4.0, Dean's List Fall 2024–Present

Relevant Coursework: Computational Physics, Mechanics (Including Orbital Mechanics), Electronics, Techniques of Experimental Physics, Modern Physics, Calculus I, II, III, Applied Linear Algebra, Physics I, II, Observational Astronomy (Astrodynamics focus)

Research & Engineering Projects

Autonomous UAV GNC Research | *Principal Investigator & Lead Engineer*

April 2026 – Present

Program Funding & Field Validation: Awarded a \$5,000 institutional grant; Partnered with Multnomah County Sheriff's Office SAR to field-test the system in an active pilot program.

- **Guidance & Path Planning:** Designed geometric guidance algorithms (rotating calipers on convex hull geometry and scanline polygon clipping) to generate terrain-optimized flight trajectories over irregular 3D environments reducing search time by 9%.
- **Navigation & State Estimation:** Integrated Bayesian probabilistic search theory and Koopman kinematic models into autonomous mission planning to dynamically update vehicle target zones based on Probability of Detection (POD) metrics.
- **Autonomy & Flight Software:** Developed a three-tier natural language command parser to translate voice inputs into flight trajectories; validated system end-to-end via 25 autonomous physical flight tests with a 96% command-execution success rate.

Rocket Avionics & Control Systems Research

June 2025 – Aug 2025

- **Avionics DAQ:** Developed and deployed an Arduino-based flight data acquisition (DAQ) payload utilizing an onboard IMU and barometric altimeter to track real-time rocket trajectories.
- **Control Loop Design:** Engineered a fully analog PID controller from discrete components, simulating circuitry in LTSpice and manufacturing a custom PCB to validate 2D state-space stabilization on a benchtop self-balancing platform.
- **Flight Data Analysis:** Conducted post-flight telemetry analysis on 20+ launch trajectories using MATLAB and Python; designed a Kalman filter to filter high-frequency structural noise and accurately reconstruct vehicle altitude and velocity profiles.

Micro-Wind Turbine Electromechanical Design

June 2023 – Sept 2023

- **Hardware Fabrication:** Designed and manufactured a functional small-scale wind turbine by retrofitting and modifying salvaged electromechanical components to generate off-grid electrical power.
- **Power Systems Integration:** Implemented safe electrical assembly, circuit fabrication, and power distribution systems to deliver reliable lighting to a rural facility in Nicaragua lacking electrical infrastructure.
- **Resource-Constrained Engineering:** Applied principles of electromagnetism, rotational fluid mechanics, and blade-element physics to achieve sustained renewable energy output under strict component and tooling limitations.

Leadership & Professional Experience

Team Leader | Multnomah County Search and Rescue – Portland, OR

Sept 2021 – Aug 2024

- Directed field operations for cross-functional teams of 10–15 individuals in high-pressure environments, emphasizing safety, communication, and rapid problem-solving.
- Logged 1,000+ operational hours as a 911 wilderness first responder locating and assisting missing or injured individuals in remote terrain throughout Oregon.
- Coordinated training and field operations as highest-ranked Team Leader, strengthening leadership, stakeholder coordination, and high-reliability teamwork.

Outdoor Gear Technician | Ahlberg Gear House – Colorado Springs, CO

March 2025 – Present

- Repair and calibrate technical outdoor equipment to ensure safe performance.
- Guide customers with trip planning, gear selection, and safety protocols.
- Manage inventory systems to track thousands of items and ensure readiness for field use.

Technical Skills

- **Programming:** Python, MATLAB, Simulink, Mathematica, C++.
- **Embedded & Robotics:** ArduPilot, MAVLink, MAVProxy, Arduino.
- **Tools & Software:** SolidWorks, LTSpice, Altium, Soldering, TheSkyX, Lab Instrumentation.
- **Languages:** Italian, Spanish (Oregon State Seal of Biliteracy).